Lab 1
Take The Lead
Product Description

Jermaine Lockett

Old Dominion University

CS 411W

October 24, 2026

Draft 1



Figure 1. Company Logo

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1 Introduction

Unexpected veterinary expenses can leave pet owners seeking financial assistance while arranging care. Finding payment options, explaining the need to potential donors, and locating reliable information can add to this burden.

Organizations such as The Pet Fund and Frankie's Friends provide assistance for veterinary care [2], [3]. Not One More Vet provides education and support focused on veterinary professional well-being [4]. These resources address needs that Take The Lead aims to bring together through fundraising, educational information, and coordination with participating veterinary hospitals.

Take The Lead is a proposed web application that helps pet owners create and share campaigns for veterinary expenses. Its goal is to make fundraising and supporting resources easier to access while promoting understanding of veterinary care and professional well-being.

2 Product Description

Take The Lead allows registered users, called Members, to create fundraising campaigns containing a Story about their pet's circumstances and veterinary needs. Unregistered users, called Guests, can view public campaigns, access resources, and contribute donations.

Campaigns include a 48-hour preparation period followed by a 30-day fundraising period. During preparation, Members assemble Story content while Take The Lead representatives contact the veterinary hospital to verify costs and coordinate payment arrangements. Members then share a campaign link with potential donors.

The intended service may support a promise-to-pay agreement or an initial payment, subject to hospital agreement and available funds. The proposed maximum initial payment is $5,000; the detailed funding rule remains to be confirmed.

At campaign closure, donations first reimburse Take The Lead for any initial payment, then pay the veterinary hospital. Members may retain any remainder or donate it to Take The Lead or another campaign. Any unpaid balance remains the Member's responsibility under the applicable payment arrangement.

The prototype will provide functional fundraising, donation recording, and chatbot assistance. Payments will be simulated, with no actual charges, advances, or payouts.

2.1 Key Product Features and Capabilities

Members create and manage fundraising campaigns containing a Story, supporting media, veterinary expense information, and donation activity. Each campaign has a shareable link. Take The Lead representatives manually contact veterinary offices to verify costs and coordinate payment arrangements.

Members and Guests can contribute during a campaign's fundraising period. The full product will process payments and distribute proceeds, while the prototype will record simulated donations without transferring actual funds.

An educational library will provide information about preventative care, veterinary services, pet insurance, financial assistance, and veterinary professional well-being. A directory will allow users to locate participating emergency and urgent-care facilities by ZIP code and view their contact information.

The AI chatbot will assist with website navigation, resource lookup, and Story grammar and clarity. It will not create campaigns or provide medical diagnoses. Members remain responsible for the accuracy of their campaign content.

2.2 Major Components Hardware and Software

Take The Lead will be accessible through a modern web browser on a desktop computer, laptop, tablet, or smartphone. Users will need an internet connection. The hosting environment will require computing resources for application requests and persistent storage for campaign and account information. The design does not require specialized hardware on the user's device.

The major software components are the frontend, backend application services, database, and external services. Figure 2 illustrates the intended architecture. It depicts the full product design rather than a claim that every shown integration has been implemented.



Figure 2. Take The Lead Major Functional Component Diagram.

The frontend will use Svelte and Tailwind CSS to provide a responsive interface for viewing campaigns, accessing resources, completing forms, and interacting with the chatbot. It will communicate with backend services over HTTPS.

The backend will manage authentication, campaigns, donation records, campaign timelines, resources, and chatbot requests. It will check user permissions, validate submitted information, and access the database. The full product will also manage payment arrangements and amounts owed; the prototype will simulate payments without charging users or transferring funds.

SQLite will store accounts, campaign, donation, veterinary facility, ZIP code, and resource data. Prisma Object-Relational Mapping (ORM) will connect the backend to the database. The chatbot will use the ODU CS LLM as its model provider.

The prototype will be deployed through Coolify on infrastructure hosted by a team member. The intended production system will run on cloud infrastructure and integrate with a payment-processing service for real transactions. For the time being in the prototype, payment processing will be simulated.

4 Glossary

The following terms describe the product and its major components.

Campaign - A fundraising effort containing a Member's Story, associated media, donation activity, and campaign information.

Compassion Fatigue - Physical, emotional, or psychological exhaustion associated with repeated exposure to suffering, trauma, or distress.

ER Emergency Room - A veterinary emergency facility that provides urgent assessment and treatment for animals, potentially including services outside standard business hours.

Guest - A non-registered user who may view campaigns, contribute through supported donation functionality, and access publicly available resources.

MDP Minimum Down Payment - An agreed initial payment amount established through a payment arrangement to help a veterinary hospital proceed with treatment. The proposed maximum amount is $5,000.

MFCD (Major Functional Component Diagram) - A visual representation of an application's primary functional components and their relationships or data flows.

Member - A registered user of the Take The Lead web application who may create and manage fundraising campaigns.

NOMV (Not One More Vet) - An organization focused on veterinary mental health, suicide prevention, and the well-being of veterinary professionals.

ORM (Object-Relational Mapping) - A software approach that maps application data objects to records in a database.

Preventative Care - Veterinary services intended to maintain animal health and reduce the likelihood of illness or complications, including wellness examinations and vaccinations.

Promise-to-Pay Agreement - A formal payment commitment between the relevant parties that specifies an agreed monetary obligation and applicable payment conditions.

Story - The written narrative, photographs, videos, and other campaign content a Member uses to explain the circumstances and purpose of a fundraising request.

TTL Take The Lead - The name and abbreviation of the web application and proposed service described in this document.

UC (Urgent Care) - A veterinary care facility that treats conditions requiring timely attention but may differ from an emergency hospital in hours, surgical capabilities, and available services.

Veterinary Hospital - A facility providing animal medical services that may participate in Take The Lead's verification and payment-coordination processes.

5 References

[1] Insurance Information Institute, "Facts + Statistics: Pet Ownership and Insurance." Accessed: Oct. 24, 2026. [Online]. Available: https://www.iii.org/fact-statistic/facts-statistics-pet-ownership-and-insurance

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[3] Frankie's Friends, "Welcome." Accessed: Oct. 24, 2026. [Online]. Available: https://www.frankiesfriends.org/

[4] Not One More Vet, "Veterinary Profession Mental Wellness." Accessed: Oct. 24, 2026. [Online]. Available: https://nomv.org/